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ACS Appl. Mater. Interfaces, **Just Accepted Manuscript** • DOI: 10.1021/acsami.0c08504 • Publication Date (Web): 26 Jun 2020

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Ultrathin Cellulose Voronoi-Nanonet Membranes Enable High-Flux and Energy-Saving Water Purification

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KEYWORDS: Voronoi-nanonets; cellulose; electrospun nanofibers; phase separation; water purification

ABSTRACT : Creating a desirable porous membrane with high-flux and energy-saving properties for the purification of water containing sub-micron sized contaminants, especially pathogenic microbes is of great significance, yet still a great challenge. Herein, we demonstrate a facile methodology to construct an innovative membrane with continuous cellulose Voronoi-nanonet structures via nonsolvent induced phase separation. This approach enables cellulose Voronoi-nanonets to tightly weld with electrospun nanofibrous substrates by controlling the solvent-nonsolvent mutual diffusion process. The resultant membranes exhibit integrated properties of

small pore size (0.23 μm), high porosity (90.7%), good interconnectivity, and ultrathin thickness (~600 nm, two orders of magnitude thinner than the conventional microfiltration membrane). As a result, the prepared membranes can effectively intercept sub-micron particles (~0.3 μm) with robust rejection efficiency (>99.80%) and ultrahigh permeation flux (maximum of 8834 L m^{-2} h^{-1}) under an extremely low driving pressure (≤20 kPa). More importantly, prominent bacterial rejection efficiency with log reduction value (LRV) of 8.0 (overcoming previous limitation of LRV <7) and outstanding antifouling function are also achieved for the membranes. The successful fabrication of such a versatile membrane may provide new insights into the development of next-generation high-performance separation materials for various applications.

INTRODUCTION

Pathogenic microbial contamination of water for food manufacture and drug processing generates a fetal threat to human health, which has been identified as the one of the greatest global safety concerns.[1, 2] This issue is highlighted by a report from New Zealand's Ministry for Primary Industries in 2013 that 38 tons whey protein was contaminated by Clostridium sporogenes, which triggered the tremendous fear about food contamination around the word.[3] In addition, serious adverse drug events caused by microbial contamination have raised from ~30 000 to >90 000 over the past few decades, as monitored by the US Food and Drug Administration.[4] Therefore, effective purification of the water is particularly significant and highly desired. Currently, general water purification methods to eliminate the hazard of microbial contamination include direct disinfection and heat treatment. However, the commonly used disinfectants (such as bleaching powder, hypochlorites, and chloramine) often lead to inevitable and undesirable secondary contamination, which are not recommended in water for food manufacture and drug processing.[1, 5, 6] As to the heat treatment involving evaporation treatment, ultrahigh-temperature treatment and pasteurization, it

can indeed remove most of the bacteria, whereas is ineffective for spore-forming bacteria and also damages the heat-sensitive components in products at the same time.[7] Alternatively, membrane microfiltration, showing advantages over the above methods in terms of no secondary pollution and mild operating temperature (generally <50 °C), has attracted increasing attention in water purification.[8] However, the available membranes are mainly prepared from polyvinylidene fluoride, polyether sulfone, and cellulose acetate by phase inversion, resulting in undesirable hydrophobic surface. As a result, these membranes always suffer from performance deficiencies of low permeation fluxes (<1000 L m^{-2} h^{-1}) and high driving pressures (>100 kPa).[9-11] Therefore, extensive efforts should be further devoted to fabricating a microfiltration membrane with high flux and low energy consumption.

Cellulose, with attractive merits of superhydrophilicity, easy accessibility, and renewability, has been demonstrated as an ideal candidate for fabricating high-performance microfiltration membranes.[12, 13] Compared with conventional hydrophobic membranes, hydrophilic cellulose-based water filtration membranes prepared from phase inversion exhibit improved fluxes (generally 1000~1500 L m^{-2} h^{-1}).[14-16] However, the further improvement of flux is severely hindered by the structural limitations involving low porosity (<80%), large thickness (>100 μm), and poor channel connectivity, which result from the unreasonable tailoring of phase inversion process.[17] These structural bottlenecks have been partially overcome by loading cellulose nanofibers on electrospun nanofibrous membranes (NFMs) with high porosity and favorable interconnectivity, by impregnation or suction filtration.[18-22] However, the functional cellulose layer, formed by the rearrangement of mutually independent nanofibers, is subjected to discontinuous structure. Consequently, the isolated cellulose nanofibers are easy to permeate into

the porous channels of NFMs, and then resulting in a declined permeation flux, which tremendously impedes the practical applications of these cellulose nanofibers/NFMs.

Herein, we present a feasible methodology for construction of highly effective and robustly water-permeable membranes with continuous Voronoi-like nanonets tightly welded on the nanofibrous substrate, for the first time, by nonsolvent induced phase separation of continuous cellulose solution. The structures of the Voronoi-nanonets are effectually designed and optimized by means of regulating the coagulation bath temperature and cellulose concentration. Attributing to such continuous cellulose Voronoi-nanonets layer, the prepared membranes demonstrate distinct structural advantages of small pore size, high porosity, ultrathin thickness, and prominent interconnectivity. Consequently, it can effectually reject sub-micron sized contaminants involving TiO_2 microparticles and *Escherichia coli* (*E. coli*) with robust permeation fluxes even under ultralow pressures, also shows desirable reusability, favorable continuous filtration performance, and superior antifouling performance.

RESULTS AND DISCUSSION

Construction of cellulose Voronoi-nanonet membranes. To construct a microfiltration membrane with both high rejection efficiency and excellent permeation flux under low operating pressures, our design is based on the following two decisive criteria: (i) a continuous network layer must provide small and uniform pores, and be tightly bonded with the NFM substrate, to effectively intercept the sub-micron sized contaminants, (ii) the membrane should possess interconnected pore channels, high porosity, and low thickness, to facilitate water transport. To satisfy the two criteria, as schematically shown in Figure 1a, a membrane with continuous Voronoi-nanonets was designed based on nonsolvent induced phase separation of continuous cellulose solution. The fabrication of the membranes began with the preparation of hydrophilic but

N,N-dimethylacetamide (DMAc) insoluble polyamide 66 (PA66) NFMs, which served as the substrates by virtue of their randomly aligned porous structures and high porosity. The detailed characterization of PA66 NFMs could be found in Figure S1 and Supplementary discussions section in the Supporting Information. Subsequently, the prepared PA66 NFMs were pre-wetted by water and then dip-coated with cellulose solution. Thus, a cellulose gel layer could be formed instantaneously at the interface between the substrate and the cellulose solution due to the water filled in the porous PA66 NFMs, which can effectively prevent the diffusion of cellulose solution to the interior of the PA66 NFM substrate.[16] After that, the membranes were transferred to water bath immediately to give rise to the fast phase separation of cellulose solution, yielding continuous Voronoi-nanonets tightly welded on PA66 membranes. This ingenious strategy has never been reported previously for preparing membranes with Voronoi-nanonets structure.

Scanning electron microscopy (SEM) images of the resulted membrane revealed that continuous Voronoi-nanonets layer firmly welded with PA66 nanofibers through cellulose nanowires (Figure 1b), which is significantly distinct from the discontinuous cellulose layer formed by the accumulation of cellulose nanofibers.[20, 23] Meanwhile, the Voronoi-nanonets exhibited extremely small pore size (Figure 1b) and high coverage (Figure S2), which enabled the final membranes to intercept sub-micron sized contaminants effectively but allow the water to permeate easily. Interestingly, the average diameter of the regenerated cellulose nanowires in continuous Voronoi-nanonets was only 42 nm (Figure S3), which is similar to that of cellulose nanofibers extracted from native cellulose.[24] However, the crystal features of native cellulose and regenerated cellulose analyzed by X-ray diffraction (XRD) are entirely different (Figure 1c). It can be seen that cellulose before regeneration exhibited three peaks at 2θ of 15.0°, 16.9°, and 22.9°, which corresponded to the (110), ($1\bar{1}0$), and (200) lattice planes of cellulose I crystalline, respectively, with crystallinity

of 76.7 %.[25-27] After being dissolved in DMAc/LiCl system and regenerated in water bath, the obtained regenerated cellulose showed a broad scattering peak around 2θ of 21.0 °, which is a cellulose II crystal type, along with crystallinity of 33.1%.[25] In addition, due to the thickness of the cellulose Voronoi-like nanonet layer is ~40 nm, the resultant membranes were ultrathin with a total thickness of ~600 nm (Figure 1d), which was almost the same as PA66 NFM substrate (Figure S4) and two orders of magnitude lower than that of most microfiltration membranes ever reported.[17, 28, 29] More interestingly, this ultrathin membrane presented excellent transparency with a high transmissivity up to 74% (Figure S5 and S6). Benefitting from the unique structural advantages of small pore size, high porosity, ultrathin thickness, and favorable interconnectivity, the prepared membranes were capable of being a desired material for high-flux and energy-saving water purification.

In order to reveal the formation mechanism of the continuous Voronoi-nanonets, a scheme illustrating the regeneration process of the cellulose Voronoi-nanonets in water bath was proposed in Figure 1e. Degreasing cotton was dissolved in DMAc/LiCl system to form homogeneous solution with extended chains through the tight binding of chloride ion to the hydroxyl of cellulose and lithium ion to the carbonyl of DMAc.[30] After adding the cellulose solution to the water (nonsolvent) bath, the hydrogen bonds between chloride ions and cellulose molecules were destroyed and the chloride ions were removed into the water, thus the electric balance of the solution system was disturbed and phase separation took place.[27] In consequence, the system evolved into polymer-rich and polymer-poor phases; then the extended cellulose chains began to concentrate and associate driven by the nucleation and growth of the polymer-poor phase. [31, 32] Finally, the cellulose was regenerated into nanowires which bonded tightly with each other through chains association to form continuous Voronoi-nanonets.

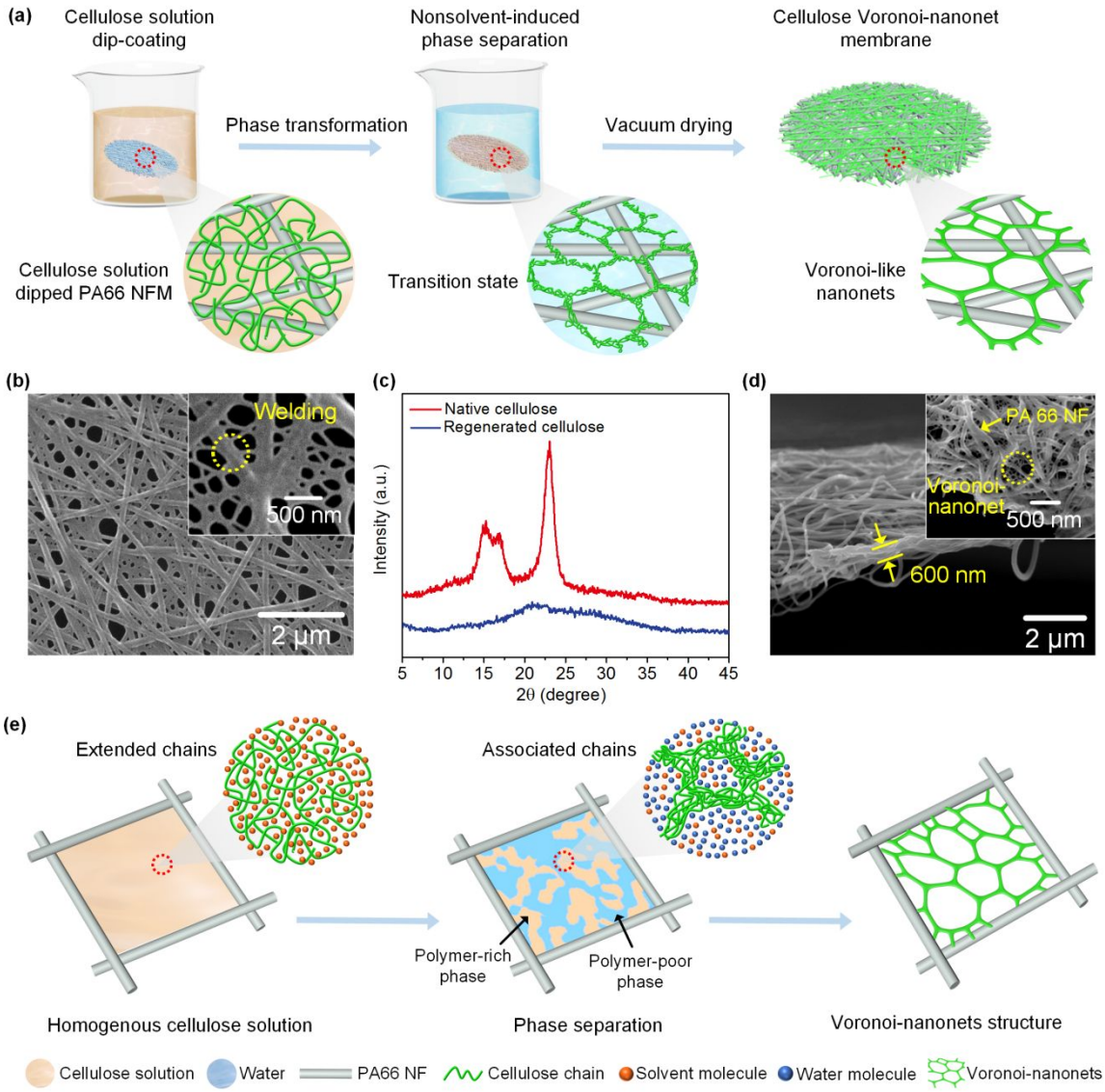

Figure 1. (a) Schematic diagram of the fabrication process of cellulose Voronoi-nanonet membrane. (b) Surface SEM images of cellulose Voronoi-nanonet membrane (M-C8 prepared from coagulation bath with temperature of 25 ^{o}C as a sample). (c) XRD patterns of cellulose before (degreasing cotton) and after regeneration. (d) Cross-sectional images of cellulose Voronoi-nanonet membrane (M-C8 prepared from coagulation bath with temperature of 25 ^{o}C as a sample). (e) Schematic illustration of the formation of Voronoi-nanonets structure.

Structural control of cellulose Voronoi-nanonet membranes. To tailor the structure of Voronoi-nanonets, coagulation bath temperature, which played a key role in controlling phase separation rate, was first regulated. The structures of cellulose Voronoi-nanonet layers prepared from various coagulation bath temperatures of 10, 25, 40, and 55 ^{o}C are displayed in Figure 2a-d. With a low temperature of 10 ^{o}C, Voronoi-nanonets along with dense films were observed, which mainly resulted from the delayed phase separation caused by the low diffusion rate between solvent and nonsolvent.[27, 33] Interestingly, when the temperature raised to 25 ^{o}C, the dense films turned into Voronoi-nanonets tightly welded on the PA66 nanofibers. Further increasing the temperature to 55 ^{o}C usually brought about reduced coverage of the Voronoi-nanonets, which could be explained from two aspects. On the one hand, the enhancement of molecules thermal motion under high temperature promoted the mutual diffusion between solvent and nonsolvent, thereby resulting in fast phase separation.[27] On the other hand, the intense thermal motion of solvent molecules caused a decreased cellulose chains/solvent interaction, leading to the morphological change of cellulose chains.[34] Consequently, it was more difficult for cellulose solution to regenerate into Voronoi-nanonets structure widely. To synthetically dissect the influence of water bath temperature on the structures of membranes, capillary flow porometer was employed to analyze the membrane pore size and pore distribution, meanwhile, a gravimetric method was used to estimate the porosity. As demonstrated in Figure 2e, along with the elevation of coagulation bath temperature from 10 to 55 ^{o}C, the average pore size of the membranes gradually increased from 0.28 to 0.40 μm, still smaller than that of the original PA66 NFM substrate (0.61 μm) (Figure S1c). Additionally, benefiting from the porous structure of the Voronoi-nanonets, the obtained membranes still showed high porosity (>90.3%) (Figure 2f), indicating a negligibly lower porosity compared with the PA66 NFM substrate (93.7%). It is worth

mentioning that, membrane prepared from coagulation bath with temperature of 10 °C showed the lowest porosity, which possibly because the formed nonporous films blocked the porous channels of the PA66 NFMs to some extent.[35] Apparently, the Voronoi-nanonets could significantly decrease the pore size of the substrate without deteriorating the porosity and pore connectivity tremendously.

The coagulation bath temperature not only affected the morphologies and pore structures of membranes, but also had an impact on the cellulose crystal structures. As shown in Figure 2g and Table S1, all the prepared membranes showed typical cellulose II crystals, but the crystallinity decreased from 37.2 to 29.3% with increasing the coagulation bath temperature from 10 to 55 °C, indicating a relatively looser structure. Although, the membranes obtained from increasing bath temperature exhibited decreased mechanical properties (Figure 2h), the optimized cellulose Voronoi-nanonet membrane was still more mechanically robust (with a tensile strength of 8.3 MPa) than PA66 NFM substrate (5.6 MPa) (Figure S1d), which may be attributed to the "nets bonding" effect from the Voronoi-nanonets.[36]

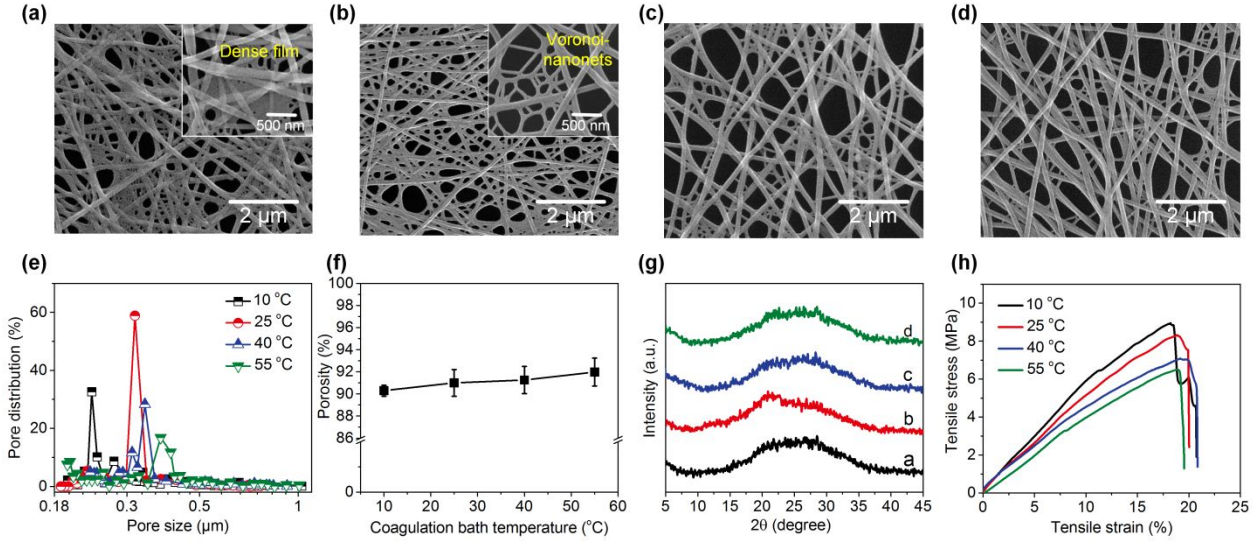

Figure 2. SEM images of the cellulose Voronoi-nanonet layers prepared from coagulation bath with temperature of (a) 10 °C, (b) 25 °C, (c) 40 °C, and (d) 55 °C (cellulose concentration of 0.004

wt%). (e) Pore size distribution and (f) porosity of the membranes obtained from the corresponding coagulation bath. (g) XRD patterns of regenerated cellulose obtained from coagulation baths with different temperatures (a. 10 ^{o}C, b. 25 ^{o}C, c. 40 ^{o}C, and d. 55 ^{o}C). (h) Tensile stress-strain curves of the membranes obtained from the corresponding coagulation bath.

Polymer concentration, as another critical parameter affecting the nonsolvent induced phase separation progress, has a significant influence on the membrane morphology.[9] Therefore, a series of membranes were prepared from dilute cellulose solutions with various concentrations (0.002, 0.004, 0.008, and 0.016 wt%) at coagulation bath of 25 ^{o}C, which were named as M-C2, M-C4, M-C8, and M-C16. As depicted in Figure 3a-d, the SEM images directly revealed the morphology change of the cellulose Voronoi-nanonets due to the variation of the solution concentration. With an increase of the cellulose concentration from 0.002 to 0.008 wt%, the coverage rate of the Voronoi-nanonets could achieve nearly 100%. However, Voronoi-nanonets structure disappeared and dense film with few pores came into sight with the cellulose concentration raised to 0.016 wt%. These results were probably related to that the varying initial cellulose concentration could change the composition path which mainly affects the final membrane morphologies (Figure 3e). Actually, the initial concentration is essentially relevant to the degree of chains entanglement, which could be reflected by solution viscosity.[37] As demonstrated in Figure 3f, the relatively low solution viscosity (≤115 mPa.s) at the initial concentration of ≤0.004 wt% implied that cellulose chains may not overlap over a large area. As a result, cellulose chains could move easily during the solvent-nonsolvent mutual diffusion process (composition path a and b in Figure 3e) and finally formed fractional Voronoi-nanonets (Figure 3g), which could not fully cover the PA66 NFM substrate. Further increasing the initial concentration to 0.008 wt%, the sharply increased viscosity indicated some entanglements emerged among cellulose chains, which could partly hinder the

mobility of the chains. Consequently, the cellulose chains associated with each other to form integrated Voronoi-nanonets structure driven by the growth of the polymer-poor phase (Figure 3h). However, excessively entangled cellulose chains at the solution viscosity of 402.5 mPa.s would in turn imped movement of the chains during the fast phase separation (composition path d in Figure 3e), resulting in undesired dense porous films (Figure 3i).[37, 38]

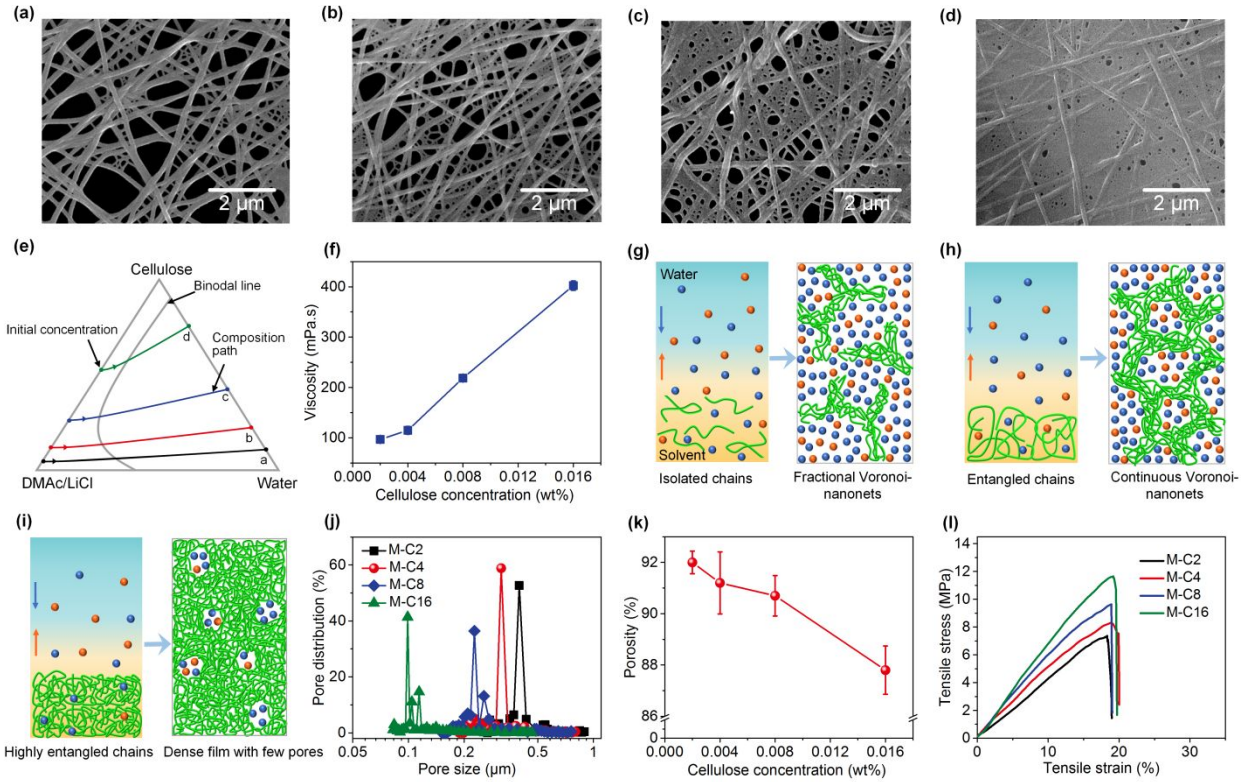

Figure 3. SEM images of the cellulose Voronoi-nanonet layers obtained from dilute cellulose solutions with concentration of (a) 0.002 wt%, (b) 0.004 wt%, (c) 0.008 wt%, and (d) 0.016 wt%. (e) Schematic phase diagram of the system cellulose-DMAc/LiCl-water showing the composition path of the cellulose solution with different concentrations. (f) The viscosity of cellulose solutions with different concentrations. Schematic illustrating the relationship between (g) isolated, (h) entangled, and (i) highly entangled cellulose chains and the structures of Voronoi-nanonets. (j)

Pore size distribution, (k) porosity, and (l) tensile stress-strain curves of the membranes obtained from cellulose solutions with different concentrations.

To give insight into the influence of cellulose concentration on the pore structure of the membrane, quantitative characterization of the membrane pore size and porosity are presented in Figure 3j and k. Membranes obtained from cellulose concentration of 0.002, 0.004, 0.008, and 0.016 wt% exhibited average pore size of 0.40, 0.32, 0.23, and 0.10 μm, respectively (Figure 3j). Obviously, compared with the PA66 NFM substrate (average pore size of 0.61 μm, pore size distribution in the range from 0.42 to 2.36 μm) (Figure S1c), the prepared membranes showed much smaller pore size and narrower pore size distribution due to the Voronoi-nanonets or dense porous film. Furthermore, attributing to the Voronoi-nanonets and dense film which only formed on the surface of the PA66 NFM substrates without deteriorating their pore structure, the porosity of the membranes is > 87.8% (Figure 3k), which is 1–2 times higher than the majority of microfiltration membranes reported by previous studies.[39-41] Moreover, benefiting from the interlinked Voronoi-nanonets which served as "net bonding", the resultant membranes exhibited more satisfactory tensile strength (7.4–11.6 MPa) than PA66 NFM substrate (5.6 MPa) (Figure 3l). Especially, the cellulose Voronoi-nanonet membrane could endure 100-cycle tensile fatigue test with a tensile strain of 5% and 100-cycle buckling fatigue test with buckling strain of 50% (Figure S7), no obvious plastic deformation was observed after the 100-cycle test mechanical tests, implying a good elasticity and excellent practicability. In addition, the influence of nonsolvents on the cellulose Voronoi-nanonet membranes was also investigated detailedly in the Supporting Information and Supplementary discussions section.

Microparticle filtration performance estimation. The fascinating nanoporous Voronoi-nanonets structure, high porosity, ultrathin thickness, and favorable mechanical properties

encouraged us to intensively investigate the microfiltration performance of the membranes. Figure 4a demonstrates the pure water fluxes of the membranes obtained from various cellulose concentrations within 60 min under a driving force of 5 kPa. It was seen that the pure water flux of each membrane slightly declined in the first 10 min and then remained almost constant, which was similar to that of the PA66 NFM substrate (Figure S8). This phenomenon could be ascribed to the compression of the membranes under the continuous pressure.[22] Despite that, the final pure water fluxes (>3143 L m^{-2} h^{-1}) of these membranes were still higher than that of conventional microfiltration membranes.[10, 42] Such robust water flux mostly benefited from the perfect membrane structures involving the high porosity, prominent connectivity, and ultrathin thickness, on the basis of the Hagen-Poiseuille equation, which was defined as follows: $J = \frac{\varepsilon \pi r^2 \Delta p}{8 \mu L}$, where J is the flux of the liquid , Δp and μ are the applied pressure and viscosity of the liquid, respectively, ε represents the porosity, r represents the pore radius, and L represents membrane thickness.[43, 44] The microparticles filtration performance was tested in the dead-end filtration apparatus by using 100 ppm suspension of TiO_2 particles (~0.3 μm) as the simulated contaminant (Figure S9 and S10). As illustrated in Figure 4b, with increasing cellulose concentration from 0.002 to 0.008 wt%, the rejection efficiency of the membrane sharply enhanced to 99.90% from 96.50% by virtue of the intriguing capture capability of the continuous Voronoi-nanonets. Further raising the cellulose concentration to 0.016 wt% resulted in almost unchanged rejection efficiency (99.93%) but a drastically declined permeation flux (form 4360 to 2401 L m^{-2} h^{-1}) (Figure 4c), which could be ascribed to that the formed dense film could effectively intercept the microparticles but reduce the water channels, and as a result caused an increasing resistance to liquid flow. Figure 4d showed the simulated water flow field within membranes with different structures. Obviously, dense film presented predominant colors of green and yellow, in dramatic contrast to the situation (blue and

cyan) of Voronoi-nanonets, confirming the contribution of Voronoi-nanonets on achieving low water pressure. This simulation, in accordance with the experimental results, further suggested that Voronoi-nanonets can be a promising alternative for high-performance water purification.

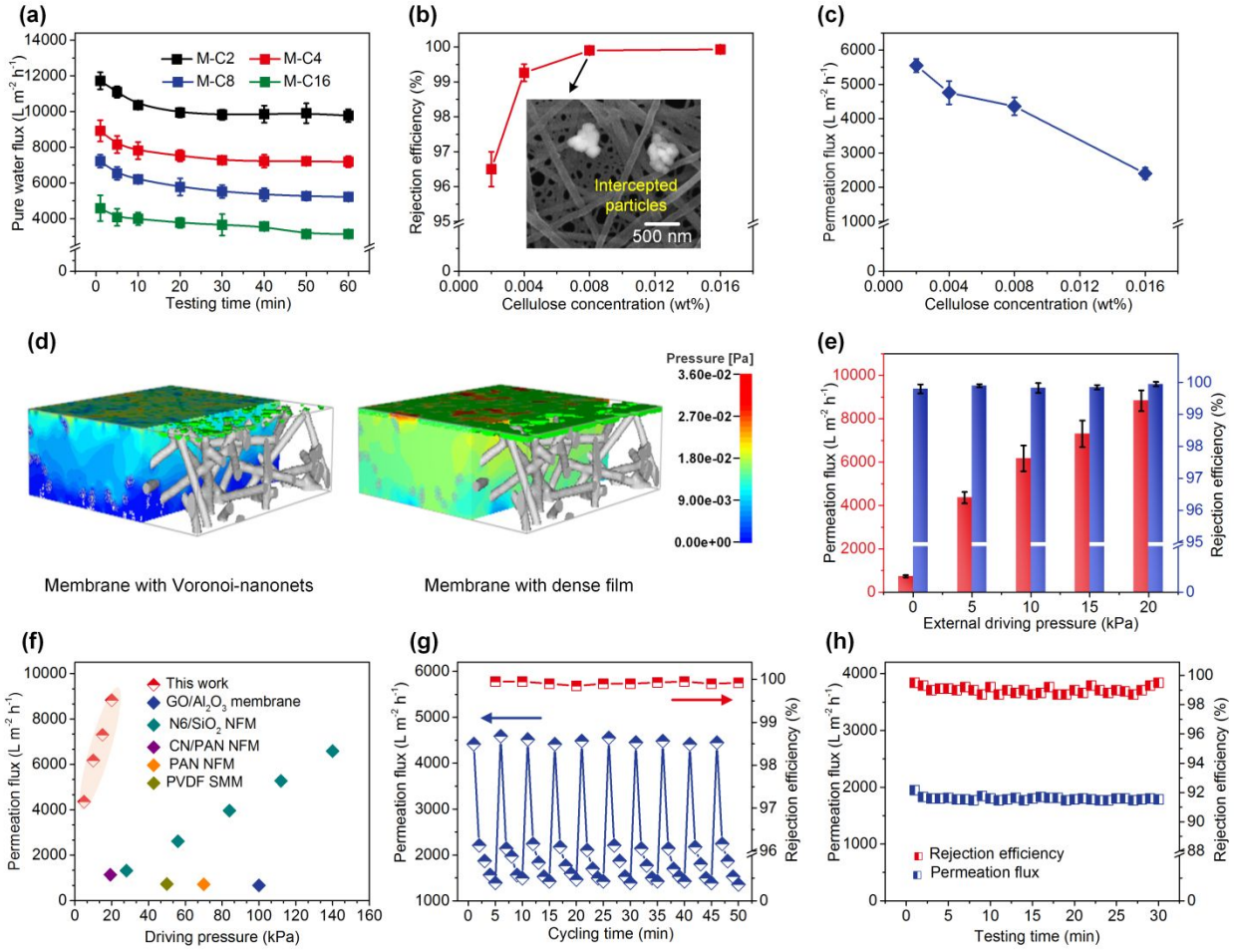

Figure 4. (a) Pure water fluxes of the membranes during the testing time with an external driving pressure of 5 kPa. (b) Rejection efficiencies (the inset is the SEM image of M-C8 after microfiltration), and (c) permeation fluxes of the membranes prepared from cellulose solutions with different concentrations (driving pressure of 5 kPa). (d) 3D models showing the pressure field while water flow penetrating through the membranes with difference structures. (e) Permeation fluxes and rejection efficiencies of M-C8 at different external driving pressures. (f) Comparison of the relative permeation flux and driving pressure between cellulose Voronoi-nanonet membrane

and the current microfiltration membranes. (g) The cycling filtration performance of M-C8 under the driving force of 5 kPa. (h) The permeation flux and rejection efficiency of M-C8 during the cross-flow filtration process.

Benefiting from the superior structures, M-C8 was chose for the estimation of a series of microfiltration experiments. It could effectually intercept the microparticles with a favorable permeation flux of 736 L m^{-2} h^{-1} and rejection efficiency of 99.80% even under the driving of gravity (Figure 4e, Movie S1 and Figure S11a). When a tiny driving pressure of 5 kPa was applied, the permeation flux distinctly raised from 736 to 4360 L m^{-2} h^{-1}. Further boosting the operating pressure to 20 kPa, the permeation flux increased up to 8834 L m^{-2} h^{-1}, while the rejection efficiency almost unchanged and still kept above 99.80%, which could be owed to the robust structure stability of the cellulose Voronoi-nanonet membrane (Figure 4e, Movie S2 and Figure S11b). Considering such excellent microfiltration performance was achieved at low driving pressure, the cellulose Voronoi-nanonet membrane was more attractive than the conventional microfiltration membranes and a series of pioneering membranes ever reported from an energy conservation point of view (Figure 4f).[18, 45-48]

Additionally, the reusability of the cellulose Voronoi-nanonet membrane was also assessed through cycling separation for the TiO_2 microparticle suspension by adopting the same dead-end filtration system. As shown in Figure 4g, the permeation flux demonstrated a sharp decline in the first minute and then a moderate decrease in the remaining time during each cycle. Such phenomenon may be ascribed to the rapid generation of a filter cake on the Voronoi-nanonets layer which seriously prevented the water permeation.[49] Surprisingly, thanks to the surface interception of continuous Voronoi-nanonets, the trapped microparticles could be easily removed by washing. Thus, the permeation flux could almost entirely recover to the initial value and filtrate with purity

over 99.90% was obtained even after 10 filtration cycles, implying the durability of the membrane. From the perspective of practical application, if the intercepted microparticles could be removed simultaneously without disturbing the purification process, a continuous and time-saving filtration would be realized. Therefore, a home-made cross-flow module connected to a peristaltic pump was adopted to estimate the continuous filtration performance (Figure S12). As expected, the membrane maintained an almost constant permeation flux (~1800 L m^{-2} h^{-1}) and high rejection efficiency (> 99%) during the whole filtration process (Figure 4h and Movie S3), suggesting a favorable practicability.

Microorganism rejection and antifouling performances evaluation. According to the standard of United States Pharmacopoeia, a sterilizing-grade microfiltration membrane should meet the requirement of intercepting the bacteria with rejection efficiency of log reduction value (LRV) $\geqslant 7$.[50] Here, *E. coli*, the typical waterborne pathogen, was used to evaluate the capture capability of cellulose Voronoi-nanonet membrane for pathogenic microbes. As illustrated in Figure 5a, compared with PA66 NFM substrate (LRV = 1.9), the cellulose Voronoi-nanonet membrane exhibited excellent bacterial rejection efficiency of LRV = 8.0, which was in accord with the standard of United States Pharmacopoeia. The drastic increase of the bacterial rejection efficiency was attributed to that the Voronoi-nanonets was capable of easily capturing the *E. coli* cells (Figure 5b). Notably, due to the appropriate pore structures (higher porosity and favorable interconnectivity) and ultrathin thickness (~600 nm), our cellulose Voronoi-nanonet membrane showed relatively higher permeation flux (1444 L m^{-2} h^{-1}) than the commercial advanced Millipore membrane (713 L m^{-2} h^{-1}), although they possessed the similar bacterial performance (LRV of cellulose Voronoi-nanonet membrane = 8.0 and LRV of Millipore membrane = 8.5).

Membrane fouling caused by microorganism adhesion is a major problem for membrane filtration, which increases the operating costs and restricts its practical application.[51] Thus, the antifouling performance of cellulose Voronoi-nanonet membrane was also examined and reflected by SYBG Green (SG) stained fluorescence images, as displayed in Figure 5c. Compared with the PA66 NFM substrate showing relatively high bacterial colonies of $(2.7 \pm 0.2) \times 10^4$ *E. coli* cells per cm^2, only $(4.9 \pm 0.2) \times 10^3$ *E. coli* cells per cm^2 could be found in cellulose Voronoi-nanonet membrane surface, indicating a more excellent antifouling property. Such obvious difference in antifouling performance may be associated with the disparate surface chemistry.[52] As depicted in Figure 5d, the abundant amide groups existing on PA66 NFM surface could easily bond with bacteria through specific adhesion, resulting in a high propensity of fouling. For cellulose Voronoi-nanonet membrane, however, the strong hydration capability of the hydroxyl groups created large hydrophilic repulsion force between bacteria and the membrane, contributing to the easy detachment of bacteria.[53, 54] As a result, cellulose Voronoi-nanonet membrane possessed more excellent antifouling properties. The cellulose Voronoi-nanonet membrane could also be employed to directly purify water by simply fixing it in a syringe filter and feeding water containing 1×10^8 CFU ml^{-1} *E. coli*, as illustrated in Figure 5e. The plate assay indicated that no bacterial colony was discovered in the collected filtrate, implying the high effectiveness of the membrane in purifying the bacteria-contaminated water. Significantly, the cellulose Voronoi-nanonet membrane could effectively intercept the bacteria with stable rejection efficiency (LRV $\geqslant$8.0) within the test filtration amount of 100 L m^{-2} (Figure 5e) , which has rarely been found in other water purification membranes.[55] Moreover, as demonstrated in Figure 5f, the microorganism rejection performance of cellulose Voronoi-nanonet membrane is obviously superior to those intensively studied materials (CN/PAN, PAN-Cs, PAN/ZnO etc.).[18, 56, 57] Considering the high

sterilization efficiency and robust permeation flux, the cellulose Voronoi-nanonet membrane could be a promising material for widespread applications.

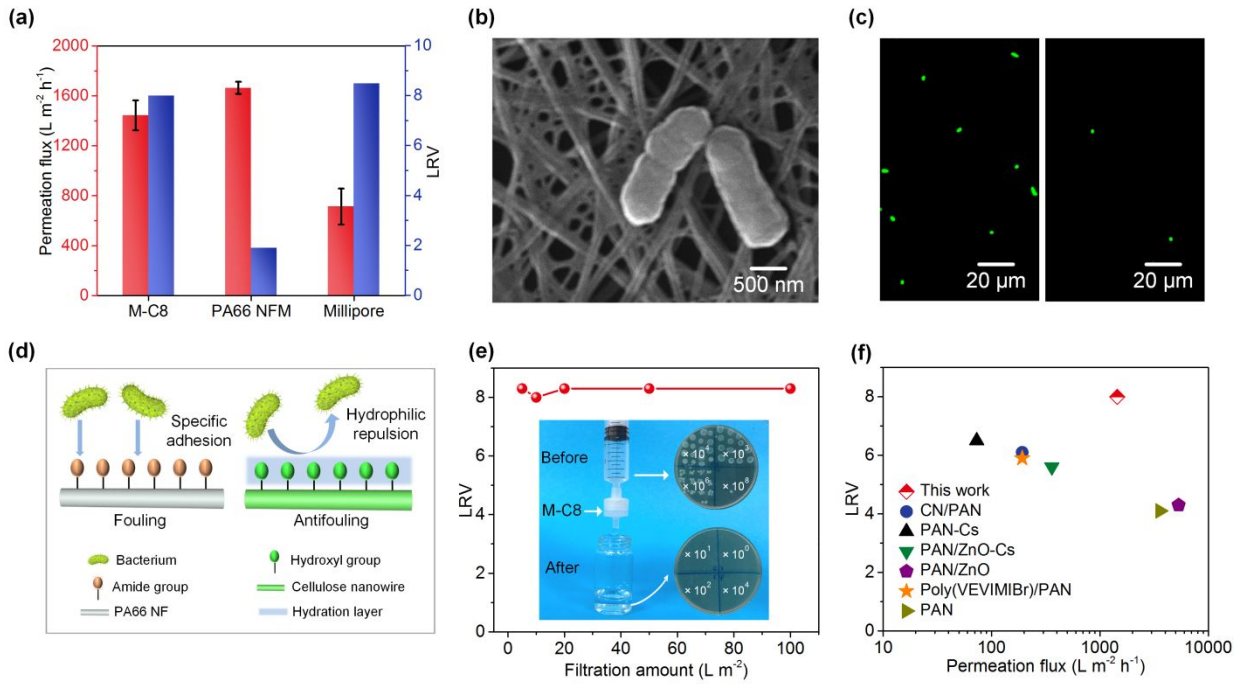

Figure 5. (a) Permeation flux and LRV against *E. coli* of M-C8, PA66 NFM, and Millipore GSWP. (b) SEM image of M-C8 after filtration of *E. coli* (at low bacterial concentration). (c) SG stained fluorescence images of PA66 NFM (left) and M-C8 (right) after incubating with *E. coli*. (d) Schematic illustration of the antifouling mechanism of cellulose Voronoi-nanonet membrane. (e) LRV of M-C8 as a function of filtration amount of *E. coli*-contained water (inset is a photograph of the filtration of *E. coli* mixture using the syringe filter). (f) LRV and permeation flux of selected microfiltration membranes.

CONCLUSIONS

In conclusion, we have successfully demonstrated a robust and facile strategy for fabricating membranes with continuous cellulose Voronoi-nanonets structure by nonsolvent induced phase separation. The structures of the Voronoi-nanonets were effectually controlled by facilely

adjusting the coagulation bath temperature and the cellulose concentration. The continuous Voronoi-nanonets endowed the membranes with superior structures involving small pore size (0.23 μm), high porosity (90.7%), favorable interconnectivity, and ultrathin thickness (~600 nm). As a result, the prepared cellulose Voronoi-nanonet membranes exhibited a robust filtration performance for ~0.3 μm TiO_2 particles with excellent rejection efficiency (>99.80%), high permeation flux (maximum of 8834 L m^{-2} h^{-1}) under low driving pressures (≤20 kPa), desirable reusability and impressive continuous filtration performance. Significantly, a permeation flux of 736 L m^{-2} h^{-1} together with excellent rejection efficiency of 99.80% was achieved merely at the driving of self-gravity (≈1 kPa), which was rarely been found in other microfiltration membranes with the same pore size. Meanwhile, the obtained cellulose Voronoi-nanonet membranes also demonstrated excellent sterilization (LRV of 8.0, and permeation flux of 1444 L m^{-2} h^{-1}) and antifouling performances. We hope that the successful construction of such cellulose Voronoi-nanonet membranes could provide new approach for designing the next-generation separation membranes which possess high performance and are capable of being applied in various applications such as medical health, bioengineering, and food processing.

EXPERIMENTAL SECTION

Materials. PA66 chips (M_w = 18 000) were produced by UBE Industries Ltd., Japan. Natural degreasing cotton was obtained from Xinxiang Piaoan Sanitary Materials Co., Ltd., China. Formic acid (>88%), DMAc (≥99.5%), ethanol (≥99.7%), and glutaraldehyde (50%) were produced by Shanghai Chemical Reagents Co, Ltd., China. LiCl (≥99%), TiO_2 microparticles (~0.3 μm), and phosphate buffer saline (PH = 7.4) (PBS) were produced by Shanghai Aladdin Chemistry Co., Ltd., China. The commercial microfiltration membrane (Millipore GSWP 0.22) was obtained from Millipore China Co., Ltd., USA. LB Lennox broth, LB Lennox agar, and SG DNA dye were

purchased from Sangon Biotech Co., Ltd., China. The model bacteria used in microorganism rejection and antifouling tests was *E. coli* (ATCC 25922, Gram-negative bacteria), which was provided by Nanjing Le Zhen Biotech Co., Ltd., China. Pure water was obtained from a Heal Force NW system.

Fabrication of electrospun PA66 NFM substrates. The precursor PA66 solution for fabrication of the substrate in this work was prepared by dissolving 14 wt% PA66 chips in formic acid under continuous mechanical agitating for 8 h at room temperate. The PA66 NFM substrate was fabricated by using DXES-N spinning machine. Typically, the prepared PA66 solution was transferred to four syringes which equipped with 6 G metal nozzles. Then the nozzles were connected with a stable voltage of 30 kV. Afterwards, the precursor solution was extruded out from the nozzles with a constant flow rate of 0.5 mL h^{-1}. The generated PA66 nanofibers were deposited on the hydrophilic polypropylene nonwoven fabrics wrapped around a metal rotating roller, with a 25 cm tip-to-collector distance. The ambient temperature during the electrospinning process was controlled at 25 ± 2 ^{o}C and the relative humidity was kept at 45 ± 5%.

Preparation of cellulose Voronoi-nanonet membranes. The nonsolvent induced phase separation technique was applied to construct Voronoi-nanonets structured membranes. Firstly, dilute cellulose solutions with different concentrations (0.002, 0.004, 0.008, and 0.016 wt%) were prepared by dissolving degreasing cotton in DMAc/LiCl (weight ratio of 1: 0.08) under continuous mechanical stirring at 100 ^{o}C for 8 h. For the formation of Voronoi-nanonets structured membranes, PA66 NFMs with size of 3 × 3 cm^2 were first immersed in pure water, where the filled water could minimize the intrusion of the cellulose solution. Then the wetted PA66 NFMs were dipped in cellulose solution and scraped by a glass rod. Subsequently, the PA66 NFMs coated with cellulose solution were immersed in water bath with temperature of 10, 25, 40, and 55 ^{o}C to

undergo phase separation. Then the wetted membranes were dried in a heating oven for 60 min under 30 °C to remove the residual water. Finally, membranes with Voronoi-nanonets layer was obtained.

Microfiltration experiments. To evaluate the microfiltration performances of prepared membranes involving the pure water permeation and microparticle interception, a commercial dead-end filtration system and a home-made cross-flow filtration system were adopted. For pure water permeation measurement, disc membranes with effective area of 2 cm^2 were fixed in dead-end filtration system connected with a vacuum pump. Then pure water was poured into the filter bowl and permeated through the membranes at a driving pressure of 5 kPa. Microparticle interception performances were investigated by dead-end filtration and cross-flow filtration. Before the tests, TiO_2 microparticles and pure water were mixed with weight ratio of 1: 9999, then the mixture was treated by ultrasonic for 30 min at 25 ^oC to obtained TiO_2 microparticle suspension with concentration of 100 ppm. For dead-end filtration, the membranes were first wetted by water to remove the air in the pore channels and then fixed in the dead-end filtration apparatus. Subsequently, TiO_2 microparticle suspension was immediately poured onto the membranes and permeated through the membranes under different external driving pressures of 0, 5, 10, 15, and 20 kPa. During this process, the suspension was kept at a nearly constant height ($\approx$ 10 cm). Furthermore, the reusability of the membrane was also evaluated *via* cycling filtration of TiO_2 microparticles for 10 cycles by employing the dead-end filtration apparatus driven by a pressure of 5 kPa, and each cycle lasted for 5 min. The membrane was rinsed by pure water to remove the intercepted microparticles after every filtration cycle. For the cross-flow filtration, the pre-wetted membrane with size of 2 × 2 cm^2 was installed in the module, then the TiO_2 microparticle suspension was pumped into the module through a peristaltic pump with a constant feed rate of 20

rpm min^{-1}, and the filtrate was collected from the hole beneath the module. The rejection efficiency of the membranes were determined by the following equation $Rejection = \left(1 - \frac{C_p}{C_f}\right) \times 100\%$, where C_f and C_p express the concentration of TiO_2 microparticles in the feed and permeation liquid, respectively.[58] The concentrations of the liquids before and after filtration were tested by UV/VIS spectrophotometer (U-3900/3900H, Hitachi, Japan). The fluxes were estimated using the equation as follows: $J = \frac{Q}{A\,(\Delta t)}$, in which J represents the liquid flux (L m^{-2} h^{-1}), Q represents volume of the permeated liquid within one minute, A is the effective area of the tested membranes, and Δt is the sampling time.[59]

Microorganism rejection and antifouling assays. Microorganism rejection tests were performed on the typical gram-negative bacteria *E. coli*. Firstly, *E. coli* was cultured in 10 mL LB Lennox broth with an initial concentration of 20 g L^{-1} at 37 ^{o}C for 24 h. Afterwards, the bacteria were collected by centrifugation at 5000 rpm for 8 min, and then repeatedly rinsed with PBS buffer solution for two times, followed by diluting in 30 mL pure water to obtained *E. coli* suspension with a concentration of 1 × 10^{8} CFU mL^{-1}. Subsequently, the bacterial suspension was filtered using the dead-end filtration system driven by a pressure of 5 kPa. The rejection efficiency for bacteria was estimated in terms of LRV defined as $LRV = log\,(\frac{C_f}{C_P})$, in which C_f represents the bacterial concentration of the feed liquid and C_P represents the bacterial concentration in the permeation liquid.[18] Meanwhile, drop plate method was applied to enumerate bacteria. Typically, the feed and permeate suspension was serially diluted and then plated on LB Lennox broth agar plates. After the drops on the agar dried, the plates were cultivated for 24 h at 37°C, and then the number of bacterial colonies was counted. For the antifouling experiment, cellulose Voronoi-nanonet membrane and PA66 NFM were first incubated in *E. coli* suspension for 24h with ambient

temperature of 25 °C. Subsequently, these membranes were rinsed with PBS buffer solution several times to get rid of the planktonic bacteria. Then the membranes were stained with SG DNA dye with a concentration of 5 ug mL^{-1}. After incubated in the dark for 20 min, these membranes were placed on a microscope slide and examined using a laser scanning confocal microscope.

Characterization. SEM (VEGA3 LMU, Tescan, Czech Republic) was applied to observe the morphologies of the membranes. Here, to observe the Voronoi-nanonets structure more clearly, SEM samples were prepared by peeling Voronoi-nanonets layer together with only one or two layers of nanofibers from the substrate (except for the samples for cross-sectional SEM images). XRD (TD-3500, Dongda, China) was used to measure the crystalline structure of cellulose before and after regeneration with a scattering angle from 5° to 45° and 2θ at a scanning speed of 2° min^{-1}. The crystallinity was calculated according to the following equation $C = \frac{I_{200} - I_{am}}{I_{200}} \times 100(\%)$, where C represents the crystallinity defined by Segal and coworkers, I_{200} gives the maximum intensity of the peak corresponding to 200 lattice plane in the sample. I_{am} expresses the minimum intensity between 110 and 200 lattice planes.[25, 26] Capillary flow porometer (CFP-1100AI, Porous Materials Inc., USA) was used to characterize the porous structures of membranes. A gravimetric method was employed to analyze the membrane porosity by the equation as follows: $\varepsilon(\%) = \frac{W_w - W_d}{\rho A L}$, in which, W_w and W_d are the weight of membrane in wet and dry state respectively, ρ represents the water density (0.998 g cm^{-3}) at 25 °C, A and L are the test area and membrane thickness.[46] Tensile tester (New Fiber Instrument XQ-1C, Shanghai, China) was employed to measure the membrane mechanical properties. Dynamic mechanical analyser (TA-Q850, TA USA) was used to test the membrane cycle tensile and buckling performance. Digital viscometer (DV3TLV, Brookfield, America) was adopted to measure the viscosity of the cellulose solutions. UV/VIS spectrophotometer (U-3900/3900H, Hitachi Japan) was used to test the transmissivity of

the membrane. Dynamic light scattering (Brookhaven Instruments BI-200SM, USA) was adopted to characterized the particle size and distribution of TiO_2 microparticles. Laser scanning confocal microscope (UltraView VoX, PerkinElmer, USA) was employed to study the antifouling properties of the membranes. Image processing software (ImageJ V1.8.0) was adopted to calculate the number of bacteria adhering to the surface of membranes according to the fluorescence images.

ASSOCIATED CONTENT

Supporting information. SEM image, fiber diameter distribution, pore size distribution, and tensile stress-strain curve of PA66 NFM, SEM image of cellulose Voronoi-nanonet membrane with low magnification, cellulose nanowire diameter distribution, cross-sectional SEM image of PA66 NFM, photograph and transmissivity of the cellulose Voronoi-nanonet membrane, cycle fatigue test of cellulose Voronoi-nanonet membrane, pure water flux of PA66 NFM, photographs of the dead-end filtration and cross-flow filtration systems, dynamic light scattering size distribution of TiO_2 mocronparticles in the feed solution, SEM images of the cellulose Voronoi-nanonet membranes prepared from different nonsolvents, crystallinity of regenerated cellulose prepared from coagulation bath with different temperature and Supplementary discussions (PDF).

Supporting movies (AVI).

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENT

This work was supported by the National Natural Science Foundation of China (no. 51973028, 51925302, and 51903036), the Shanghai Committee of Science and Technology (no. 19ZR1401200), the Fundamental Research Funds for the Central Universities (no. LZB2019004, and CUSF-DH-D-2017017), Huo Ying Dong Education Foundation (no. 171065), and the Director Fund of State Key Laboratory for Modification of Chemical Fibers and Polymer Materials (no. ZR1902).

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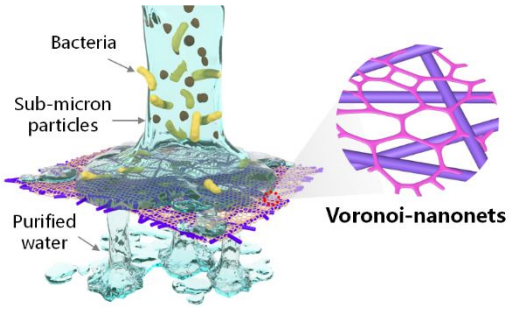
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